

1.3 Volumetric Analysis

1. (a) Number of moles of iodine reacted = $c \times V$

$$= 0.0250 \times 0.01425$$

$$= 3.56 \times 10^{-4} \text{ mol}$$

$$n_{\text{iodine}} : n_{\text{vitamin C}} = 1 : 1$$

$$\text{Therefore } n_{\text{vitamin C}} \text{ in } 25.0 \text{ mL} = 3.56 \times 10^{-4} \text{ mol}$$

(4 marks)

$$C = \frac{n}{V} = \frac{3.56 \times 10^{-4}}{25.0 \times 10^{-3}} = 0.0142 \text{ mol L}^{-1} \text{ (3sf)}$$

- (b) number of moles of vitamin C in 100 mL

$$= c \times V$$

$$= 0.100 \times 0.0142$$

$$= 0.00142 \text{ mol}$$

$$\text{mass}_{\text{vitamin C}} = n \times M$$

$$= 0.00142 \times 176 = 0.251 \text{ g;}$$

$$\text{(ie 0.125 g per tablet)}$$

(3 marks)

$$\% \text{ vitamin C} = \frac{0.251}{0.500} \times \frac{100}{1} = 50.2\% \text{ (3sf)}$$

- (c) The contents of the flask turn from colourless to blue-black. (For colour changes always: Colour *before* then colour *after* the reaction.) (2 marks)

2. Transfer the solid sodium thiosulfate into the volumetric flask using a clean, dry funnel. Thoroughly rinse the funnel and the beaker which held the solid with distilled water, ensuring that all rinsing water goes into the volumetric flask. Add about 150 mL of distilled water to the volumetric flask and shake the mixture with the lid on until the solid has fully dissolved. Fill the volumetric flask to just below the calibration mark with distilled water. Top up the solution to the calibration mark one drop at a time using a teat pipette until the bottom of the meniscus is level with the line. Replace the lid and shake the mixture thoroughly. (7 marks)

3. (a) no. of moles of thiosulphate = $c \times V$
 $= 0.100 \times 0.01275$
 $= 0.00128 \text{ mol (3 sig. fig.)}$

(1 mark)

- (b) $n_{\text{iodine}} = n_{\text{thiosulfate}} = 1 : 2$
 Therefore $n_{\text{iodine}} = 0.5 \times 0.01275$
 $= 0.000638 \text{ moles}$

(1 mark)

- (c) Mole ratio $\text{I}_2 : \text{ClO}^-$ is 1 : 1
 Therefore number of moles of hypochlorite ions = 0.000638 mol

(1 mark)

- (d) mass of NaOCl = $n \times M$
 $= 0.0006375 \times [22.99 + 35.45 + 16.00]$
 $= 0.0006375 \times 74.44$
 $= 0.04746 \text{ g}$

(2 marks)

$$\text{Conc}^n \text{ of NaOCl} = \frac{0.0475}{0.0200} = 2.37 \text{ g L}^{-1}$$

- (e) Conc
- ⁿ
- of the undiluted bleach

$$= 2.37 \times \frac{250}{5.00} = 119 \text{ g L}^{-1}$$

(2 marks)

This is equivalent to 11.9 g per 100 mL

- (f) Chlorine gas can escape from the bottle of bleach during storage or during the titration, lowering the concentration of the 'available chlorine'. (1 mark)

- (g) (i) This would decrease slightly the number of moles of bleach pipetted, so the final concentration of diluted bleach would be lower than its correct value. Hence the resultant titre value would be lower than the correct value. (2 marks)

- (ii) This would mean that the bleach would not have been diluted by the extent suggested, which would give a concentration higher than the correct value. Hence the resultant titre value would be higher than the correct value. (2 marks)

4. (a) moles of HCl =
- $c \times V = 0.0100 \times 0.0100 = 1.00 \times 10^{-4} \text{ mol}$
- (2 marks)

- (b) moles of NaOH =
- $c \times V = 0.0100 \times 0.00523 = 5.23 \times 10^{-5} \text{ mol}$

This reacts with 5.23×10^{-5} moles unreacted HCl (as indicated by the 1:1 mole ratio in the equation). (3 marks)

- (c) moles of HCl which reacted with NH
- ₃
-
- $= 1.00 \times 10^{-4} - 5.23 \times 10^{-5} = 4.77 \times 10^{-5} \text{ mol}$
- (2 marks)

- (d) (i)
- 4.77×10^{-5}
- moles HCl reacted with
- 4.77×10^{-5}
- moles of NH
- ₃
- (as indicated by the 1:1 mole ratio).

Concentration of ammonia in 100.0 mL effluent

$$= \frac{n}{V} = \frac{4.77 \times 10^{-5}}{0.1000} = 4.77 \times 10^{-4} \text{ mol L}^{-1}$$
 (3 marks)

- (ii)
- $4.77 \times 10^{-4} \text{ mol L}^{-1}$
-
- $= 4.77 \times [14.01 + 3(1.008)]$
-
- $= 8.13 \times 10^{-3} \text{ g L}^{-1}$

In 1000 L there would be $8.13 \times 10^{-3} \times 1000 \text{ g} = 8.13 \text{ g}$ Therefore concⁿ = 8.13 g per 1000 L = 8.13 (ppm) (3 marks)

- (e)
- $(10.0 \text{ mL HCl} - 5.23 \text{ mL NaOH}) \times 1.703 = 4.77 \times 1.703 = 8.12 \text{ ppm NH}_3$
- (2 marks)

- (f) Prediction: smaller titre value.

Explanation: If the amount of ammonia in the effluent increases then more hydrochloric acid will react in Step 2 and more ammonium chloride will be formed. This will leave less unreacted hydrochloric acid present to react with the sodium hydroxide in Step 3, lowering the titre. (3 marks)

- 5 (a)

$$n = \frac{m}{M} = \frac{5.25}{74.44} = 7.05 \times 10^{-2} \text{ mol}$$

$$C = \frac{n}{V} = \frac{7.05 \times 10^{-2}}{0.1} = 0.705 \text{ mol L}^{-1}$$

(2 marks)

- (b) 705

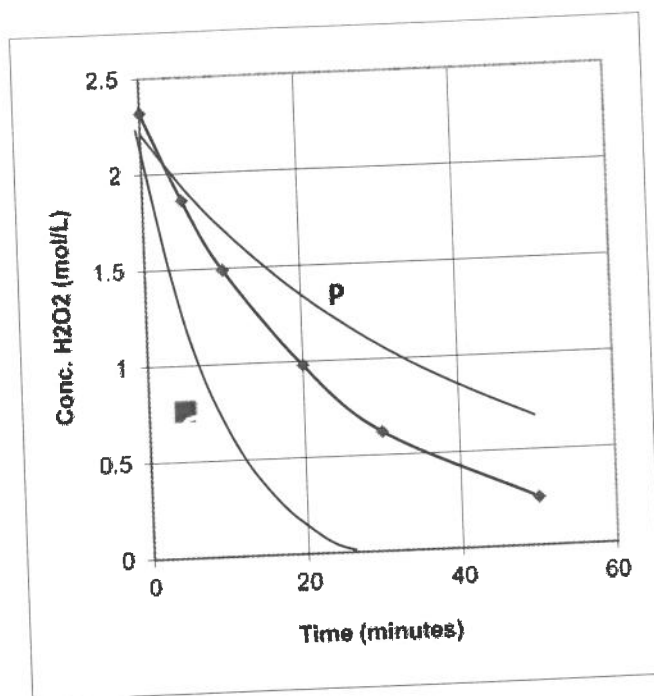
(1 mark)

- (c) Mole ratio of NaOCl : Cl₂ is 1:1
Therefore 705 moles of chlorine needed
 $m = n \times M = 705 \times 70.9 = 5.00 \times 10^4 \text{ g}$ (3 sig. figs) (2 marks)
6. (a) [Any *one* of the following:]: To ensure that the concentration is the same as the original wine. Greater accuracy if use a diluted sample. Harder to recognise the end-point of concentrated solution. Volume of dichromate used could be excessive. (1 mark)
- (b) $n = c \times V$
 $= 1.25 \times 0.0200$
 $= 0.0250 \text{ mol}$ (3 sig fig.) (2 marks)
- (c) $\text{Cr}_2\text{O}_7^{2-} + 6\text{Fe}^{2+} + 14\text{H}^+ \rightarrow 2\text{Cr}^{3+} + 6\text{Fe}^{3+} + 7\text{H}_2\text{O}$ (1 mark)
- (d) (i) Moles of $\text{Cr}_2\text{O}_7^{2-} = 6.91 \times 10^{-3}$.
Mole ratio $\text{Fe}^{2+} : \text{Cr}_2\text{O}_7^{2-} = 6:1$
Moles of $\text{Fe}^{2+} = 6 \times 6.91 \times 10^{-3}$
 $= 4.146 \times 10^{-2}$
 $V = \frac{n}{C} = \frac{4.146 \times 10^{-2}}{2.00} = 2.07 \times 10^{-2} \text{ L}$
 $= 20.7 \text{ mL}$ (3 sig. fig.) (3 marks)
- (ii) Moles of $\text{Cr}_2\text{O}_7^{2-}$ reacted
 $= 0.0250 - 0.00691$
 $= 0.01809 \text{ mol}$
 $= 0.0181 \text{ mol}$ (3 sig fig) (1 mark)
- (iii) Mole ratio $\text{Cr}_2\text{O}_7^{2-} : \text{C}_2\text{H}_5\text{OH} = 2:3$
Moles of $\text{C}_2\text{H}_5\text{OH} = 0.01809 \times \frac{3}{2}$
 $= 0.0271 \text{ mol}$ (3 sig fig) (2 marks)
- (iv) $c = \frac{n}{V} = \frac{0.027135}{0.01}$
 $= 2.71 \text{ mol L}^{-1}$ (3 sig fig) (1 mark)
- (v) 0.0271 mol in the 10.0 mL sample
Therefore 0.271 mol in 100.0 mL sample (wine $\equiv$ diluted distillate)
Mass of ethanol = $n \times M$
 $= 0.271 \times 46.068$
 $= 12.5 \text{ g}$ (3 sig fig) (2 marks)
- (vi) No heating means that less oxidation is going to occur. If less oxidation is going to occur then more dichromate is unreacted. Therefore a greater volume of Fe^{2+} is going to be needed. (3 marks)

1.4 Chromatography

1. The two compounds move up the thin layer at different rates, carried along by the mobile phase (solvent).
The rate of movement is affected by factors such as their solubility in the solvent and the extent to which they are attracted to the stationary phase.
These factors are influenced by the number of sites which can form hydrogen bonds with the mobile phase or the stationary phase
A contains *more polar* groups than **B** and is therefore adsorbed more strongly onto the

4. (a)



(6 marks)

(b) (i) See graph labelled P.

(1 mark)

(ii) See graph labelled Q.

(1 mark)

(c) The concentration of the solution decreases during the reaction. According to the collision theory, reactions occur due to successful collisions between reactant particles. As the concentration of particles is initially higher, the frequency of collisions will be greater than after 25 minutes (when much of the H₂O₂ has decomposed) at which time the rate of the reaction will be less since less collisions exceed the activation energy.

(3 marks)

2.2 Equilibrium and Yield

1. (a) If the pressure of the reaction mixture is increased, this favours a shift in the equilibrium in such a way as to lower the pressure to counteract the change (Le Châtelier's principle). This will cause a "move to the right" as this produces a smaller number of moles of gas (1 mole compared to two moles on the left). Consequently a high pressure will increase the yield of ethanol.

(3 marks)

(b) (1)

$$K_c = \frac{[\text{CH}_3\text{OH}_{(g)}]}{[\text{CO}_{(g)}][\text{H}_{2(g)}]^2}$$

(1 mark)

(2) This shows that as the temperature increases the yield of methanol decreases.

(1 mark)

(3) As temperature increases yield decreases. Therefore the reverse reaction must be endothermic (as the lowest yield is produced by the highest temperature). Therefore the forward reaction must be exothermic.

(3 marks)

- (4) Too high a temperature (eg T_3) will reduce the yield of methanol.
 Too low a temperature (eg T_1) will cause the reaction to proceed too slowly.
 Therefore a moderate temperature (eg T_2) is chosen as a compromise of these factors. (2 marks)

2. When the pH of wine is increased the concentration of $H^+_{(aq)}$ ions decreases.
 This causes a shift in the equilibrium of the second reaction "to the right" to counteract this change, which lowers the concentration of sulfurous acid, $H_2SO_{3(aq)}$.
 The decreased concentration of this acid causes the first reaction to "move to the right", lowering the concentration of aqueous sulphur dioxide, $SO_{2(aq)}$. (4 marks)

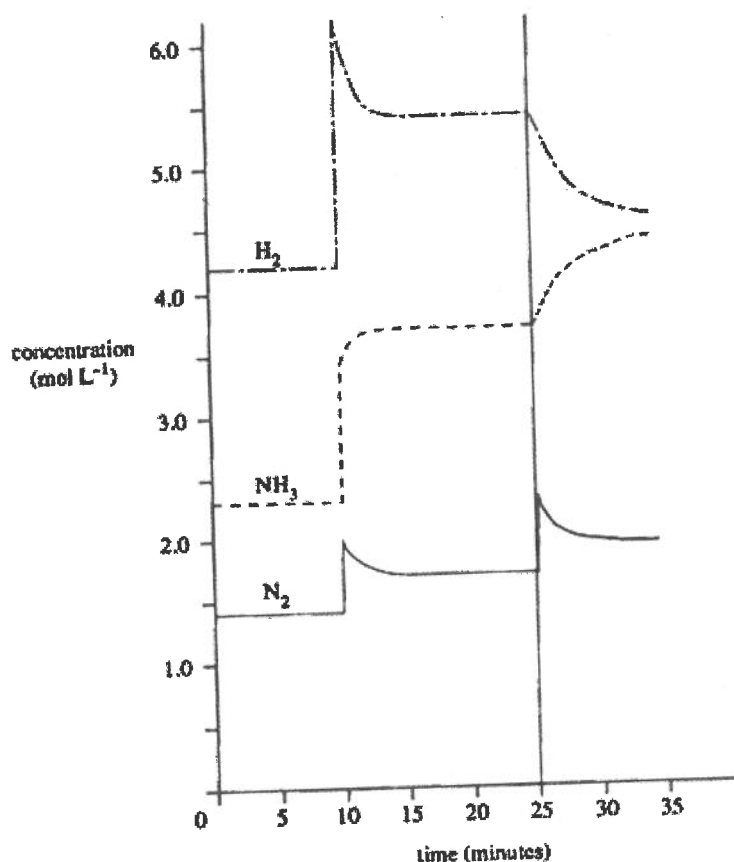
3. (a) $[N_2] = 1.4 \text{ mol L}^{-1}$ $[H_2] = 4.2 \text{ mol L}^{-1}$ $[NH_3] = 2.3 \text{ mol L}^{-1}$

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{2.3^2}{1.4 \times 4.2^3} = 0.051 \quad (5 \text{ marks})$$

- (b) At the 10-minute mark the pressure of the system was increased, by decreasing the volume of the reaction vessel.
 This caused a sudden increase in the concentration of all three substances. (ie system under stress and therefore will act to relieve the stress, of Le Châtelier's principle).
 Since the forward reaction produces fewer molecules of gas than the reverse reaction the forward reaction will be favoured to counteract the pressure increase.
 This will cause the concentration of ammonia to increase and the concentration of both reactants to decrease. (4 marks)

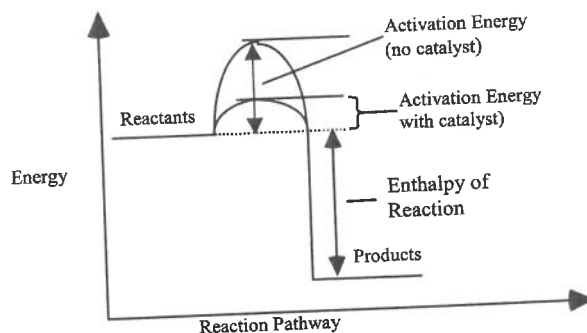
- (c) All concentrations were constant at the 15 minute mark. (1 mark)

(d)



(4 marks)

4. (a) $2\text{NO}_2 + 4\text{CO} \rightleftharpoons \text{N}_2 + 4\text{CO}_2$. (2 marks)
- (b) The porous material increases the surface area of the catalyst thereby increasing the rate of reaction. (1 mark)
- (c) $2\text{NO} + 2\text{CO} \rightleftharpoons \text{N}_2 + 2\text{CO}_2$. (2 marks)
- (d)



- (e) (i) The rate of reaction increases. (3 marks)
- (ii) $K_c = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]}$. (1 mark)
- (iii) $K_c = \frac{(1.4 \times 10^{-3})^2}{(3.5 \times 10^{-2})(8.9 \times 10^{-3})}$ (2 marks)
- $= 6.3 \times 10^{-3}$ (2 sig figs)
- (iv) Equilibrium only occurs in closed systems. Exhaust gases are not in a closed system (1 mark)

5. (a) $K_c = \frac{[\text{NO}_2]^2}{[\text{NO}]^2[\text{O}_2]}$. (2 marks)
- (b) (i) $= \frac{[0.01]^2}{[0.05]^2[0.08]} = 0.50 \neq K_c$ Therefore not at equilibrium. (3 marks)
- (ii) Increased T causes decrease in K_c . Reaction gone to the left. Reverse reaction is endothermic. Increase in T favours endothermic reaction. Forward reaction is exothermic. (3 marks)



	2HI	$\rightleftharpoons$	I ₂	H ₂
Initially	55×10^{-4}		0	0
Change	-12×10^{-4}		$+6.0 \times 10^{-4}$	$+6.0 \times 10^{-4}$
Equil ^m	43×10^{-4}		$+6.0 \times 10^{-4}$	$+6.0 \times 10^{-4}$

The initial concentration of HI = $55 \times 10^{-4} \text{ mol L}^{-1}$ (2 marks)

- (b) The intensity of the purple colour of the mixture would remain constant when equilibrium was established. (1 mark)

$$(c) \quad K_c = \frac{[H_2][I_2]}{[HI]^2}$$

(1 mark)

$$(d) \quad K_c = \frac{(6.0 \times 10^{-4} \times 6.0 \times 10^{-4})}{(43 \times 10^{-4})} = 0.0195 = 0.020 \text{ (2 sig figs)}$$

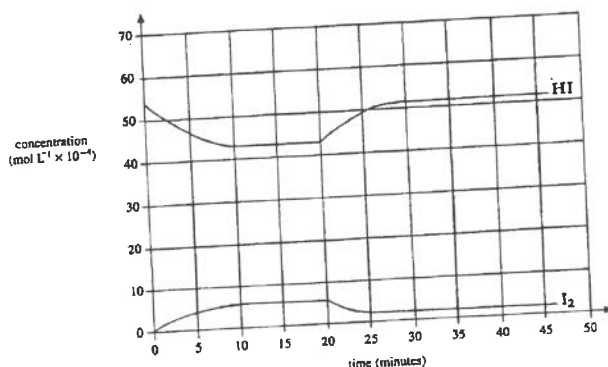
(2 marks)

(4 marks)

(e) See graph below

(3 marks)

(f) See graph below



(1 mark)

7. (a) 9 minutes.



Initial Conc. 7 0

React/Form. -4 +1

Equil^m 3 1Equil^m concn = 1 mol.L⁻¹.

(2 marks)

(c) The graph must:

- start at 0 minutes
- reach equilibrium at the same time as your answer in (d) (1)
- stop at 11 minutes

(3 marks)

Shape very important

(1 mark)

8. (a) T₂ is higher.

(b) **Explanation:** An increased temperature causes the reaction to move to the right (more moles of CO formed at the higher temperature T₂ – see graph). Increased temperature causes the reaction to occur to decrease temperature. Therefore the reaction occurs in the endothermic direction (moles of CO increase as temperature increases T₂ > T₁).
Sign of enthalpy change: Positive.

(3 marks)

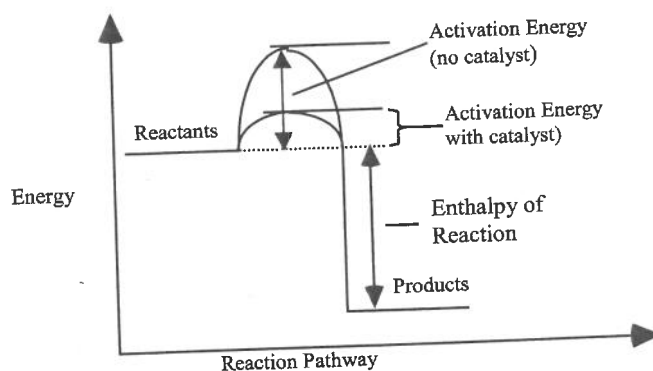
$$(c) \quad K_c = \frac{[H_2]^3 [CO]}{[CH_4] [H_2O]}$$

(1 mark)

$$(d) \quad K_c = \frac{(0.84)^3 (0.75)}{(0.25)(0.10)} = 18 \text{ (2 sig figs)}$$

(2 marks)

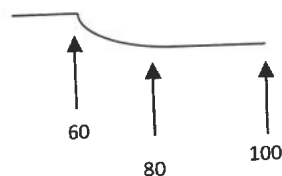
9. (a) Decreases. (1 mark)
- (b) Increase in temperature gives less SO_3 . Therefore the reaction occurs to the left. But increasing the temperature causes the reaction to occur to decrease the temperature. Therefore the reaction to the left is endothermic and the forward reaction is exothermic. (2 marks)
- (c) Reaction reaches equilibrium faster. (1 mark)
- (d) Heat reactants. (1 mark)
- (e) (i) Increase surface area. (1 mark)
- (ii) Decreases. (1 mark)
- (iii)



(2 marks)

- (f) (i) Removal/loss of SO_3 . Stress on equilibrium. Response to counter stress. Formation of more SO_3 with corresponding decrease in $[\text{SO}_2]$ and $[\text{O}_2]$. Equilibrium restored after 10 min. (4 marks)

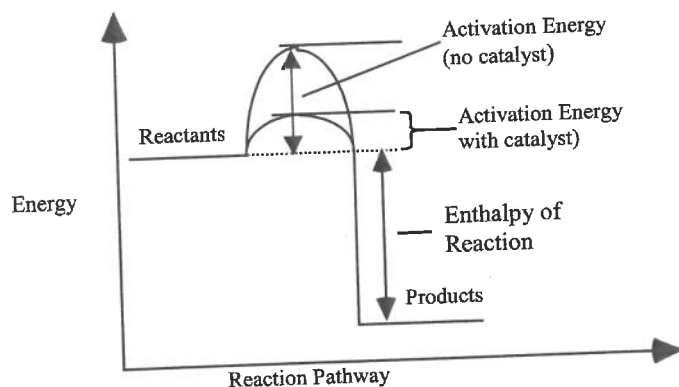
(ii)



(3 marks)

2.3 Optimising Production

1. (a) (i)



The use of a catalyst allows a different reaction pathway which has a lower activation energy required for the reaction.

Ensuring that a greater proportion of the reactant particles have energies greater than the new, lower activation energy; i.e. sufficient energy for a successful collision, thus increasing reaction rate. (3 marks)

(ii) N_2 , H_2 , CO_2 (3 marks)

(iii) The energy that is removed as heat in the condenser could be used to heat other stages in the process, eg in the shift converter. (1 mark)

(iv) natural gas, steam and air. (3 marks)

(v) carbon dioxide. (1 mark)

2. (a) $\text{NaCl} + \text{NH}_3 + \text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{NaHCO}_3 + \text{NH}_4\text{Cl}$ (2 marks)

(b) sodium hydrogencarbonate (or sodium bicarbonate) (1 mark)

(c) carbon dioxide (1 mark)

3. (a) Sea water, chlorine (2 marks)

(b) Water molecules are polar and bromine molecules are non-polar. Bromine molecules cannot overcome the extensive hydrogen bonding between water molecules resulting in two immiscible layers. (2 marks)

(c) Sodium chloride, calcium chloride, magnesium chloride etc. (any one of a chloride salt). (1 mark)

(d) Production and transfer of heat requires the combustion of fuels and generation of electricity which is expensive to the manufacturer. Combustion of fuels and electricity production results in the release of greenhouse gases (carbon dioxide) to the troposphere, which further contribute to global warming. (2 marks)

(e) Chlorine gas is a by-product of the process. The distillation of chlorine gas from sea water can be directly input into the reaction column. This reduces the amount, and cost, of chlorine gas as a raw material for the process. (3 marks)

- (f) To improve efficiency heat exchangers can be used to collect steam and generate electricity for the plant. (2 marks)
4. (a) (i) Atomic absorption spectroscopy. (1 mark)
- (ii) 1 L produces 1.29 g
 x L produces 10^6 g
 $x = \frac{10^6}{1.29} = 775193.7$ L
 $= 775\,000$ L (3 sig figs) (2 marks)
- (iii) Carbon dioxide. (1 mark)
- (iv) [Any **two** of the following]. (2 marks)
- Heater
 - Driers
 - solar evaporators
 - electrolysis units
- (v) $\text{Mg}^{2+} + 2\text{OH}^- \rightarrow \text{Mg}(\text{OH})_2$ (2 marks)
- (vi) If water is present, magnesium ions *and* water are at the cathode. Water is more easily reduced, therefore no magnesium forms. (Magnesium being an active metal can only be produced from electrolysis of a molten solution). (2 marks)

Topic 3: Organic and Biological Chemistry

3.1 Introduction

- 1 (a) pent-2-ene. (2 marks)
- (b) 3-methyl butanoic acid. (2 marks)
2. (a)
- $$\text{HO}-\text{CH}_2-\text{CH}_2-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$$
- (2 marks)
- (b)
- $$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{CH}-\text{C}-\text{CH}_2\text{CH}_3 \\ | \quad | \\ \text{CH}_3 \quad \text{CH}_2 \\ | \\ \text{CH}_3 \end{array}$$
- (2 marks)
3. (a)
- $$\begin{array}{c} \text{O} \\ || \\ \text{CH}_3-\text{C}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3 \end{array}$$
- (2 marks)
- (b) 2,2,3-trimethylpentane (2 marks)